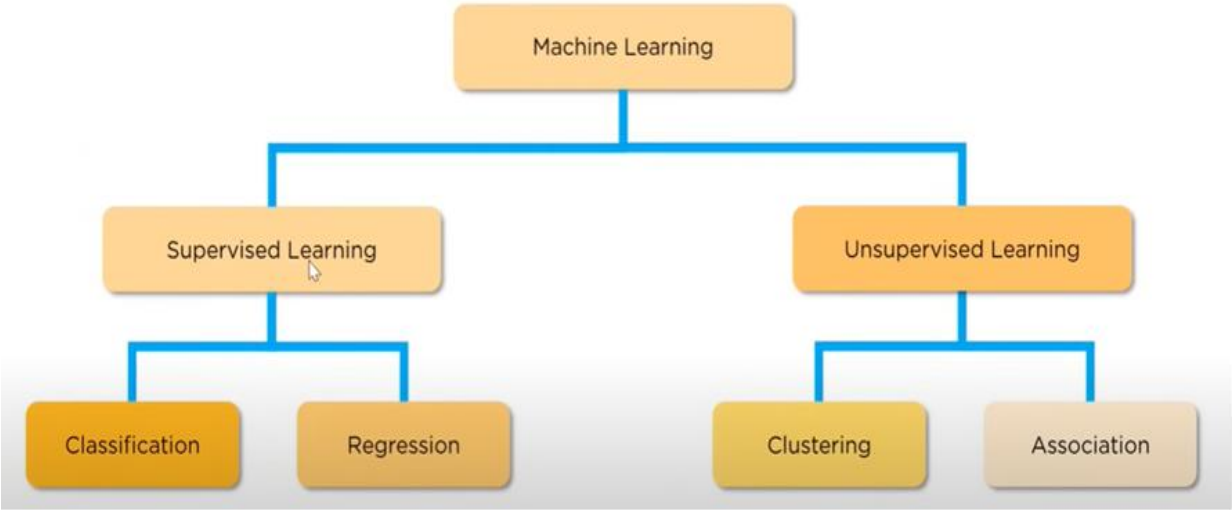
Linear Regression in Machine learning

Machine Learning is a branch of Artificial intelligence that focuses on the development of algorithms and statistical models that can learn from and make predictions on data. **Linear regression** is also a type of machine-learning algorithm more specifically a **supervised machine-learning algorithm** that learns from the labelled datasets and maps the data points to the most optimized linear functions. which can be used for prediction on new datasets.

First of we should know what supervised machine learning algorithms is. It is a type of machine learning where the algorithm learns from labelled data. Labeled data means the dataset whose respective target value is already known. Supervised learning has two types:

- **Classification**: It predicts the class of the dataset based on the independent input variable. Class is the categorical or discrete values. like the image of an animal is a cat or dog?
- **Regression**: It predicts the continuous output variables based on the independent input variable. like the prediction of house prices based on different parameters like house age, distance from the main road, location, area, etc.



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What is Linear Regression?

Linear regression is a type of supervised machine learning algorithm that computes the linear relationship between the dependent variable and one or more independent features by fitting a linear equation to observed data.

When there is only one independent feature, it is known as Simple Linear Regression, and when there are more than one feature, it is known as Multiple Linear Regression.

Similarly, when there is only one dependent variable, it is considered Univariate Linear Regression, while when there are more than one dependent variables, it is known as Multivariate Regression.

Why Linear Regression is Important?

The interpretability of linear regression is a notable strength. The model's equation provides clear coefficients that elucidate the impact of each independent variable on the dependent variable, facilitating a deeper understanding of the underlying dynamics. Its simplicity is a virtue, as linear regression is transparent, easy to implement, and serves as a foundational concept for more complex algorithms.

Linear regression is not merely a predictive tool; it forms the basis for various advanced models. Techniques like regularization and support vector machines draw inspiration from linear regression, expanding its utility. Additionally, linear regression is a cornerstone in assumption testing, enabling researchers to validate key assumptions about the data.

Types of Linear Regression

There are two main types of linear regression:

Simple Linear Regression

This is the simplest form of linear regression, and it involves only one independent variable and one dependent variable. The equation for simple linear regression is:

$y=\beta 0+\beta 1X$

where:

- Y is the dependent variable
- X is the independent variable
- β0 is the intercept
- β1 is the slope

Multiple Linear Regression

This involves more than one independent variable and one dependent variable. The equation for multiple linear regression is:

$y=\beta 0+\beta 1X+\beta 2X+\ldots\ldots\ldots\beta nX$

where:

- Y is the dependent variable
- X1, X2, …, Xp are the independent variables
- β0 is the intercept
- β1, β2, …, βn are the slopes

The goal of the algorithm is to find the best Fit Line equation that can predict the values based on the independent variables.

In regression set of records are present with X and Y values and these values are used to learn a function so if you want to predict Y from an unknown X this learned function can be used. In regression we have to find the value of Y, So, a function is required that predicts continuous Y in the case of regression given X as independent features.

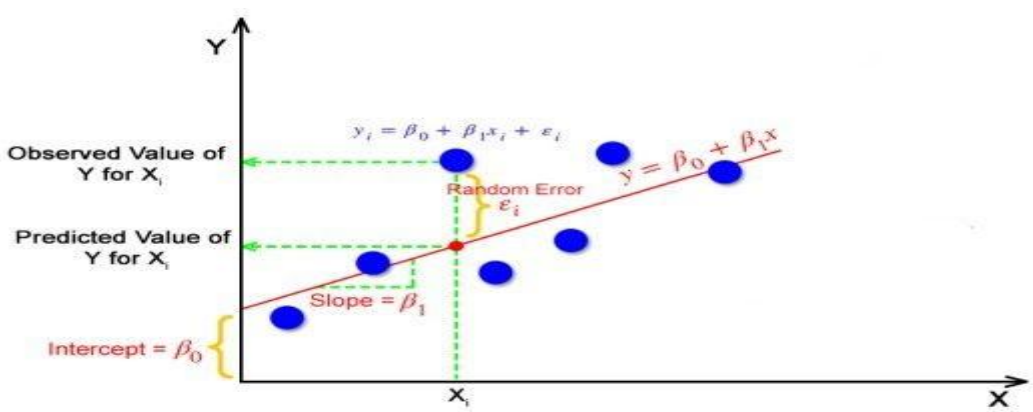
Simple Regression Calculation

To calculate best-fit line linear regression uses a traditional slope-intercept form which is given below,

$$\mathbf{Y_i = \beta_0 + \beta_1 X_i}$$

where Y_i = Dependent variable, $\boldsymbol{\beta_0}$ = constant/Intercept, $\boldsymbol{\beta_1}$ = Slope/Intercept, $\mathbf{X_i}$ = Independent variable.

This algorithm explains the linear relationship between the dependent(output) variable y and the independent(predictor) variable X using a straight line $Y = B_0 + B_1 X$.



But how the linear regression finds out which is the best fit line?

The goal of the linear regression algorithm is to get the **best values for B_0 and B_1** to find the best fit line. The best fit line is a line that has the least error which means the error between predicted values and actual values should be minimum.

Random Error(Residuals)

In regression, the difference between the observed value of the dependent variable(**y_i**) and the predicted value(**predicted**) is called the residuals.

$\varepsilon_i = y_{predicted} - y_i$

where $y_{predicted} = B_0 + B_1 X_i$

What is the best fit line?

In simple terms, the best fit line is a line that fits the given scatter plot in the best way. Mathematically, the best fit line is obtained by minimizing the Residual Sum of Squares(RSS).

Cost Function for Linear Regression

The cost function helps to work out the optimal values for B_0 and B_1, which provides the best fit line for the data points.

In Linear Regression, generally **Mean Squared Error (MSE)** cost function is used, which is the average of squared error that occurred between the $y_{predicted}$ and y_i.

We calculate MSE using simple linear equation y=mx+b:

$$MSE = \frac{1}{N}\sum_{i=1}^{n}(y_i - (B1x_i + B0))^2$$

Using the MSE function, we'll update the values of B_0 and B_1 such that the MSE value settles at the minima. These parameters can be determined using the gradient descent method such that the value for the cost function is minimum.

Mathematics Behind Simple Linear Regression

The crux of Simple Linear Regression is to derive a **best-fit line** that captures the relationship between any single independent variable and the dependent variable. The best-fit line emphasizes minimizing the overall distance of data points from the **regression line**, ensuring a precise alignment with the trend.

Statistically, the **mathematical equation** that approximately models a **linear relationship** between a dependent variable x, & an independent variable y is

$$y=mx+c$$

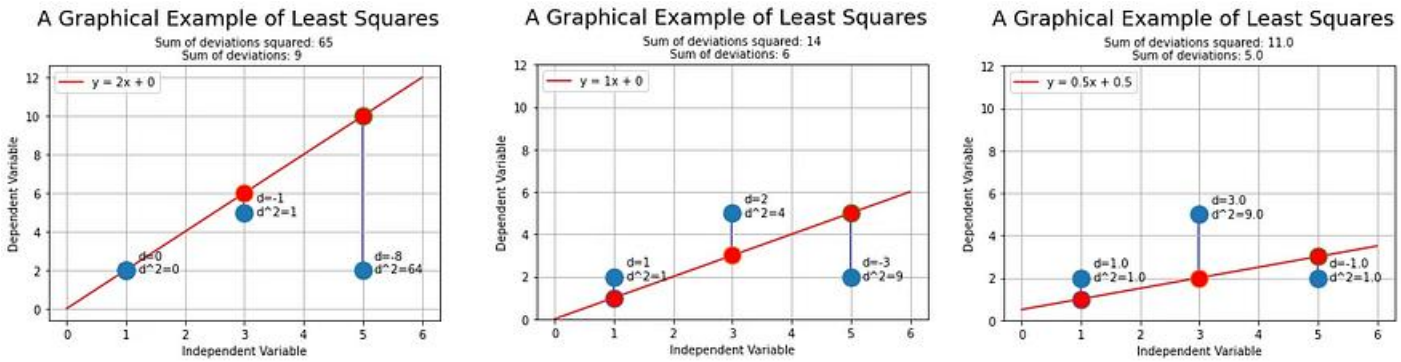
wherein, 'm' is the **slope** of the line, and 'c' is the **intercept**, and together they are referred to as the '**Model's Coefficients**'. This equation is the basis for any Linear Regression model and is often referred to as the **Hypothesis Function**

The goal of most machine learning algorithms is to construct a **model** i.e. the **hypothesis** (H) to estimate the dependent variable based on our independent variables such that it minimizes the **Loss Function (**Residual Sum of Squares)

Aptly termed the **Least Squares Method**, this approach seeks to find the most optimal values for the model parameters that minimize the loss function/RSS.

1.2 Exploring the Cost Functions

While **coefficients** provides insights into the **relationship between variables**, we need **metrics** to quantify how accurately these linear models captures the underlying data patterns. The most common metrics are RMSE and R2 Score



One of the primary metric used to assess the model's performance is **MSE,** or **Mean Squared Error**. This metric quantifies the squared difference, between the actual observed values and the predicted values from the regression line. A lower MSE indicates that the model's predictions are closer to actual values.

The **R-squared (R2) Score**, also known as the **coefficient of determination** is another vital metric that provides a measure of how well the model explains the variability, in the dependent variable. It ranges from 0 to 1, with a higher value indicating a better fit without explicitly accounting for overfitting data

$$\text{MSE}=\frac{1}{2m}\Sigma(h_\theta(x)^{(i)}-y^i)^2$$

Beyond these, you may also explore other metrices such as: **Residual Sum of Squares,** and **Residual Analysis** that offer a more comprehensive evaluation of model performance.